MILITARY SPECIFICATION

COMPASS, GYRO, VEHICLE

1. SCOPE

1.1 This specification covers the technical design and performance requirements for a direct reading and transmitting vehicle gyro compass to be used on U. S. Navy ships, vehicles, and amphibious craft.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	-Shock Tests, H.I. (High-Impact): Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-I-983	-Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.
MIL-D-1000/2	-Drawings, Engineering and Associated Lists.
MIL-Q-9858	-Quality Program Requirements.
MIL-E-17555	-Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016).

3. REQUIREMENTS

3.1 Preproduction sample.- When specified (see 6.1), a preproduction sample of the vehicle gyro compass shall be furnished for the tests in 4.3.

3.2 Description.- The vehicle gyro compass equipment shall consist of the following or equivalent configuration:

(a) Compass unit.
(b) Compass control and power inverter unit.
(c) Electronic power converter unit, when required (see 6.1).

3.3 General features.- The equipment shall be in accordance with MIL-I-983 in addition to the requirements specified herein. Whenever a requirement of MIL-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.3.1 Material.- All parts except those necessary in electromagnetic circuits, ball bearings, shafts, thrust plates and similar application where strength or hardness is essential, shall be constructed of nonferrous, nonmagnetic materials or corrosion-resisting steel.

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3.3.2 Tapping and machining.- Tapping and machining operation tolerances of less than 0.001 inch shall be completed after the corrosion-resisting treatment is applied.

3.4 Design features.-

3.4.1 Operator protection.- The equipment design shall safeguard the operator from electrical shock hazards while operating or making normal adjustments on the equipment. All electrical test points necessary for fault isolation during trouble shooting shall be accessible without disassembly of the equipment.

3.4.2 Mechanical shock and vibration.- The equipment reliability and performance accuracy shall not be adversely affected by the vibration, shock, and acceleration environments which may be expected aboard the class of vessels for which it is intended. All units shall pass the type A shock requirements of MIL-S-901 and the vibration requirements of MIL-I-983 for vital equipment. When shock and vibration mounts are required they shall be internally situated. The mechanical durability and design of the shock mount components shall provide them with a minimum life expectancy of 2 years when the equipment is installed in any location on the ship. The compass heading error due to compliance of the shock mounts shall not exceed 0.2 degrees.

3.4.3 Transistorization.- Solid state semiconductor elements rather than electron tubes shall be used in the electronic design of the equipment.

3.4.4 Enclosure.- All major units of the equipment shall be watertight in accordance with MIL-I-983.

3.4.5 Interchangeability.-

3.4.5.1 The equipment subassemblies shall be of modular construction using internal cabling with connector terminations to facilitate replacement of modules. All required interconnecting cabling, connectors and stuffing tubes shall be supplied with each equipment.

3.4.5.2 Major sub-assembly components such as servo motors, synchros, transducers, and gyro rotor and case assembly shall be designed for ready substitution of replacement units.

3.4.5.3 The Compass unit shall operate as specified when any compass control and power inverter unit is substituted without prior modification, calibration or extensive adjustment.

3.4.6. Equipment size.- The weight and dimensions of the compass equipment shall be as small as practicable and shall not exceed the limits given in this specification. The combined weight and volume of the compass equipment units shall not exceed 120 pounds and 2.5 cubic feet respectively.

3.4.7 System reliability.- The compass system shall have a predicted reliability of 4000 hours mean time between failure (MTBF) when operating in accordance with this specification in the shipboard environment. A failure consists of any malfunction which impairs the accuracy and usefulness of the compass system.

3.5 Detailed features.-

3.5.1 Compass unit.- The compass unit shall contain a gyroscopic device which automatically seeks the local reference meridian plane and continously indicates the direction of true North.

3.5.1.1 The compass shall have damped settling characteristics consistent with optimum performance accuracies under dynamic shipboard conditions. When operating in the temperature range of 20°F. to 149°F., the compass shall not require more than 4 hours for complete settling when started with an azimuth offset of 10 degrees and a tilt of approxmiately 5 arc minutes. A fast settling mode shall be employed if necessary to insure that when started under the above conditions, the compass will settle in less than 25 minutes to within 4 degrees of its defined settling point. The preceding shall apply for either static or dynamic large angle Scorsby test motions.

3.5.1.2 When a flotation fluid is employed, the tolerances on the weight and balance of the sensitive element shall be controlled to insure interchangeability of elements. The design shall preclude problems such as oil leaks, improper fluid

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temperature regulation, damage caused by fluid solidification and excessive internal pressure due to fluid expansion. The physical and chemical characteristics of the fluid shall be specified in detail. Substitute fluids and supply sources shall be identified.

3.5.1.3 The gyro rotor bearings shall be of the highest grade obtainable to insure a trouble free life expectancy of 10,000 hours.

3.5.1.4 The spin motor which drives the gyro wheel shall operate on polyphase 400 cycle electrical power. The motor shall start automaticlly when electrical power is applied. The motor performance shall be consistent with compass accuracy requirements under dynamic acceleration environments existing on craft for which the compass is intended. A motor overspeed of up to 15 percent shall not impair the operational reliability of the compass or its parts.

3.5.1.5 The compass unit shall operate satisfactorily within the limits of allowable error under all combinations of static and dynamic roll and pitch angles up to 45 degrees in amplitude.

3.5.1.6 The compass unit shall not exceed 70 pounds in weight and 1.5 cubic feet in volumes. When the compass is a "top reading" type, its overall height shall not exceed 10 inches. When the compass is a "side reading" type, its overall height shall not exceed 15 inches.

3.5.1.7 The compass shall be a direct reading and transmitting type having a minimum card diameter of 8 inches. The compass card shall have 1 degree markings in white agninst a black background. When internally illuminated during darkness, the card markings and index shall appear as red markings against a black background. The intensity of illumination shall be controlled by dimmer to a maximum luminance of 0.25 foot lamberts or more.

3.5.1.8 When the card reading index is located aft of the card, the sequence of card markings and its direction of rotation shall agree with the steering sense of the Helmsman's wheel. Card reading errors resulting from parallax shall be held to a minimum. The transparent cover through which the card is read shall not concentrate the reflection of external light sources to an extent which would interfere with comfortable reading of the card.

3.5.1.9 Means shall be provided for adjusting the heading orientation of the compass unit in the angular range of plus or minus 5 degrees. Slotted mounting holes are acceptable for this purpose. The compass unit shall have scribe lines located forward and aft to function as alignment references.

3.5.1.10 The compass unit shall be suited for operation in all geographic latitudes up to 70 degrees. The control and damping circuitry shall include the facility for simple modificatication which would extend the compass usefulness up to 80 degrees latitude.

3.5.1.11 The compass unit shall be designed for deck mounting and for intergral mounting atop the compass control and power inverter unit. Provisions for the attachment of eyebolts for the purpose of hoisting and transporting the compass unit shall be included.

3.5.1.12 Normal operation of the compass including startup and settling shall not require access to any internal elements of the equipment. External controls shall be provided for slewing and leveling the sensitive element. When provided, the caging control shall be external and readily accessible. The caged or uncaged condition shall be displayed on a visual indicator.

3.5.1.13 The follow up system shall function satisfactorily at turning rates up to 100 degrees a second. A malfunction of the follow up system shall produce a visual alarm indication.

3.5.1.14 The compass unit shall not be damaged by power interruptions and shall continue to perform with dependable heading indications immediately after power interruption intervals lasting 8 to 10 seconds under small angle scorsby conditions.

3.5.1.15 The compass design shall include the facility and capability for simple installation of size 15 synchros operating at 1 speed or 1 and 36 speed and 60 or 400 cycle synchro excitation.

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3.5.2 Compass control and power inverter unit.-

3.5.2.1 This unit shall contain the compass electronic control circuitry and operating controls necessary for startup and continous operation of the equipment. Provision shall be made for visual alarm indication of power and follow-up failure. Visual indication of caged condition shall be included. Relay controlled normally closed contacts having a current rating of 5 amperes shall be provided for the purposes of a remote ship's alarm system. The illumination of the visual indicators shall not meet dark adaptation requirements and shall also be visible under daylight conditions at distances up to 10 feet.

3.5.2.2 The design shall incorporate a recessed panel or equivalent for the mechanical protection of exposed components such as switch toggles, knobs, dials and indicators.

3.5.2.3 The magnitude and sense of compass tilt during starting and running shall be indicated continously on a meter at the control unit when tilt readout is not provided at the compass unit.

3.5.2.4 The compass control power inverter unit shall not weigh more than 30 pounds nor exceed 1/2 cubic foot in volume. The unit shall be designed for bulkhead and deck mounting. The top of the unit housing shall have mounting provisions for attachment of the compass unit.

3.5.2.5 The power inverter shall be an ungrounded static all solid state type whose output shall provide the necessary 3 phase 400 cycle power required for operation of the compass. It shall also provide electrical excitation for a 15CX4 synchro and an additional 50 watts (0.8 p.f.), 400 cycle auxilliary power to operate a remote servo type repeater. The amplitude, frequency and waveform of this auxilliary power shall permit accurate operation of the synchro and repeater load.

3.5.2.6 The power inverter shall be designed for satisfactory operation with d.c. power inputs in the range of 23 to 28 volts and not exceeding 300 watts. The input power shall be dervied from either a battery supply or from the converter unit.

3.5.2.7 The power inverter shall be designed for continuous operation. It shall have maximum practical efficiency and a minimum life of 40,000 hours. Provision shall be made to suppress input voltage spikes or switching transients of a maximum of 800 volts to a safe value required for protection of internal circuit elements and semiconductors.

3.5.2.8 The compass performance accuracy shall agree with the tolerances specified in 3.5.4.1 and 3.5.4.6 when the inverter input varies randomly in the range of 23-28 volts d.c.

3.5.3 Converter unit.-

3.5.3.1 The converter unit shall contain the solid state electronic components and circuits necessary for the conversion of 1-phase, 115 volts, 60-cycle or 400-cycle input to 24 volts d.c. output required for the operation of the power inverter.

3.5.3.2 The compass shall conform to the accuracy requirements of 3.5.4.1 and 3.5.4.5 when the converter unit input power variation is within the following tolerances: 115 volts plus or minus 10 percent, 60-cycle or 400-cycle frequency plus or minus 5 percent.

3.5.3.3 The converter unit shall not weigh more than 30 pounds or exceed 1/2 cubic foot volume space.

3.5.3.4 A power-on indicating light shall be provided on the converter unit.

3.5.3.5 The power converter unit shall be bulkhead mounted but shall also include portability features for easy handling when it is not mounted permanently.

3.5.3.6 The power converter shall be an ungrounded static type having designed maximum efficiency. It shall be designed for a minimim life of 40,000 hours and shall have circuit protection features including short circuit and overload protection, as well as the suppression of transients to insure protection of internal circuit elements.

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3.5.4 Operation.-

3.5.4.1 The following data, as applicable to the equipment units shall be supplied by the contractor prior to delivery of the first shipment of each mark and modification of the contract or order:

(a)	Maximum power	Watts
(b)	Maximum current	Amperes
(c)	Power factor	Units
(d)	Heat dissipation in each component	Watts
(e)	Average damping percentage shall be	Percent
(f)	Average compass period	Minutes

3.5.4.2 The equipment shall operate, except as specified in 3.5.4.3 within the accuracy limits specified in 3.5.4.5 at any ambient temperature between the limits of plus 20°F. and 149°F. Materials used shall conform to standard Navy requirements for classes A, B, and C insulation in accordance with MIL-I-983 and hot spot temperatures shall not exceed standard Navy established limits for these classes of insulation in each specific application.

3.5.4.3 At temperatures between minus 35°F. and plus 20°F. and between 110°F. and 149°F. twice the turntable error shall be allowed.

3.5.4.4 The equipment shall not suffer permanent damage when stored at any temperature from minus 80°F to 180°F. and shall be capable of meeting the requirements of 4.9.10.2 for exposed locations.

3.5.4.5 The limits of allowable error shall be as shown in table I. All limiting error values indicated are maximum allowable overall errors including settling point error as specified in 4.7.4.

Table I - Limits of allowable error.

Name of test	Limiting values (error in degrees)
	(Plus or minus)
Voltage and frequency variation	3
Turntable	2
Small angle Scorsby	3
Large angle Scorsby	4
Effect of magnetic fields	4
Settling point (error)	1.5

3.5.4.6 When the equipment is operated from a 24 volt d.c, or 115-volt, 1-phase, 60 or 400 cycle supply (see 3.5.3), the equipment operating conditions and accuracy shall be within the limits specified in 3.5.4.1 and 3.5.4.5, for a random variation between 23 and 28 volts in the 24-volt d.c. supply or for an oscillatory random variation in the 60 or 400 cycle, 1-phase, 115-volt supply of plus or minus 10 percent in voltage and plus or minus 5 percent in supply frequency.

3.6 Drawings.- Drawings shall conform to MIL-D-1000/2 and the detailed requirements of MIL-I-983.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirements.- The supplier shall provide and maintain a quality program acceptable to the Government. The program shall be in accordance with MIL-Q-9858. The supplier is in no way relieved of the final responsibility to furnish compasses meeting the requirements of this specification.

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4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, and (b) Government product inspection to measure quality of product to be offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 General inspection.- The procedures for the various examinations and tests to be performed on the equipment shall be as specified hereinafter. The attention of the manufacturer is invited to the applicable paragraphs of MIL-I-983 for information concerning these tests. All error limits are plus or minus magnitudes and are maximum allowable overall errors including settling point errors.

4.3 Preproduction inspection.- Preproduction inspection shall be conducted when specified (see 6.1). Retests and new samples will be required when the basic design of the equipment or the material of a vital part has been changed. The preproduction sample shall be subjected to the examination and tests specified in table II, which shall be performed in the order listed.

4.3.1 At the discretion of the Naval Ship Engineering Center (NAVSEC), a preproduction model of the equipment will be subjected to a 6 month evaluation by the U.S. Navy to determine its suitability for shipboard use.

Table II - Preproduction inspection.

Inspection	Requirement reference	Test reference
General examination	MIL-I-983	4.9.1
Dielectric strength	MIL-I-983	MIL-I-983
Insulation resistance	MIL-I-983	MIL-I-983
Roll and pitch clearance	3.5.1.5	4.9.7
Starting characteristics	3.5.1.1	4.9.4.1
Damping	3.5.4.1	4.9.3
Turntable	3.5.4.5	4.9.4
Small angle Scorsby	3.5.4.5	4.9.5
Large angle Scorsby	3.5.4.5	4.9.8
Maximum rate of turn	3.5.1.13	4.9.10.4
Voltage and frequency variation	3.5.4.5	4.9.2
Power failure	3.5.1.14	4.9.10.3
Shielding and radio frequency noise reduction	MIL-I-983	MIL-I-983
Effect of magnetic field	3.5.4.5	4.9.6
Reliability	3.4.7	4.9.10.5
Temperature and humidity	MIL-I-983	4.9.10
Enclosures	3.4.4 and MIL-I-983	MIL-I-983
Salt spray	MIL-I-983	MIL-I-983
Shock and vibration	3.4.2	4.9.9

4.4 Parts inspection.- All vehicle gyro compass equipment covered in the contract or order shall be subjected to parts inspection at the place of manufacture, as specified by the Government inspector. The Government inspector may at all times witness the manufacture of all parts of the equipment. The manufacturer shall furnish all information, apparatus, and facilities necessary to make such preliminary tests on completed units as will permit the Government inspectors to judge whether they are ready for the quality conformance inspection specified in 4.5.

4.5 Quality conformance inspection.- All equipment offered for delivery on the contract or order shall be subjected to the examination and tests specified in table III, which shall be performed, in general, in the order listed.

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Table III - Quality conformance inspection.

Inspection	Requirement reference	Test reference
General examination	MIL-I-983	4.9.1
Dielectric strength	MIL-I-983	MIL-I-983
Insulation resistance	MIL-I-983	MIL-I-983
Damping	3.5.4.1	4.9.3
Turntable	3.5.4.5	4.9.4
Small angle Scorsby	3.5.4.5	4.9.5

4.5.1 Details as to the adjustments made while the equipment is being given test runs shall be recorded for the information of the Government inspector. The equipment shall then be ready for inspection, and shall thereafter be required to pass the entire quality conformance inspection, or in case of failure, the required parts, without any changes or adjustments affecting the operation of the compass.

4.6 Special test equipment.-

4.6.1 Small angle tests require that the compass unit be placed on a Scorsby test stand. The Scorsby test stand is a device for testing gyro compasses. It is capable of subjecting the compass unit to rolling, pitching and yawing motion. The motions are independent of each other and are separately controlled as to amplitude and period. It is mounted on a turntable by means of which the heading of the compass unit may be varied at will.

4.6.2 The turntable test requires the use of a turntable, which may be incorporated in the Scorsby test stand or which may be a separate testing unit of a design suitable for producing the required movements and indicating the heading with the required degree of accuracy.

4.7 Test installation procedure and definitions.-

4.7.1 Except where specifically indicated, all tests shall be performed at room temperature (nominally 75°F.) with normal supply voltage (115 volts plus or minus 10 percent) and frequency (400-cycles plus or minus 5 percent). All readings shall be recorded in degrees to the nearest tenth. Unless otherwise stated, readings of compass indications for all tests shall be taken at 15-minute intervals. For each test the compass shall be aligned to a reference heading on the test stand, which reference heading shall be accurate to within plus or minus 30 seconds absolute. The compass shall be aligned to azimuth to the reference heading within plus or minus 3 minutes.

4.7.2 Compass errors are to recorded continuously during the tests by means of servo type strip chart recorders operating on the electrical outputs of the compass synchro transmitter. Periodic readings of the dial estimated to the nearest tenth of a degree are to be entered on data sheets.

4.7.3 Test stand heading.- The test stand heading is defined as the angle between the "zero" or forward mark on the test stand and the North point of the local meridian plane established by accurate geodetic survey.

4.7.4 Settling point.- The average of the settled positions of all headings run during the turntable test shall be used as the compass settling point. The settled condition which is a requisite for conducting ensuing compass tests is defined as being the time at which the error becomes and subsequently remains less than the allowed settling point error for a period of one hour.

4.8 Settling readings.-

4.8.1 The compass unit shall be settled prior to each test except as specified in 4.8.2. During settling readings the compass unit shall be kept stationary and level on the Scorsby test stand.

4.8.2 At the beginning of any of the tests, if the last readings for the preceding test shows that the compass unit has settled as specified in 4.7.4 the settling readings may be waived for the test being started, unless settling readings are required by the contractor's representative. However, the settling readings shall not be continued beyond the time at which the compass unit has first settled in accordance with 4.7.4.

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4.8.3 All dial readings shall be recorded to the nearest tenth of a degree.

4.9 Inspection procedures.-

4.9.1 General examination.- The general examination shall be conducted as specified in MIL-I-983. The operating test is not required as part of the general examination.

4.9.2 Voltage and frequency variation.- The compass unit shall be settled in accordance with 4.7.4 on any reading selected by the Government inspector. The voltage and frequency of all power supply to the gyro compass equipment shall then be suddenly changed to the lower limits specified in 3.5.4.6 and the compass unit shall be allowed to settle under this new condition of operation. During this settling period, readings shall be taken at 15-minute intervals. This test shall then be repeated for the higher limit specified in 3.5.4.6. A test shall then be run for random variations of the voltage and frequency of the supply voltage.

4.9.3 Damping test.- The compass unit shall be settled in accordance with 4.7.4, on any heading designated by the Government inspector. The sensitive element shall then be caged and slewed 30 degrees from the established meridan, uncaged, and leveled so that the reading is slightly off from the settled condition in such direction as to increase the deflection from the established meridan. The compass unit shall then be allowed to settle in accordance with 4.7.4. During the test, readings shall be recorded at 5-minute intervals, except on the peaks, when the interval shall be 1-minute. The compass unit shall settle as defined above within 4 hours from the time of the first peak of oscillation.

4.9.3.1 The mean percentage of damping shall not differ more than plus or minus 15 percent from the value given by the manufacturer in accordance with 3.5.4.1. The mean compass period shall not vary more than plus or minus 10 minutes from the designed period.

4.9.4 Turntable test.- The compass unit shall be mounted on a Scorsby test stand or suitable turntable, and shall be settled in accordance with 4.7.4. It shall then be given a 2-hour test on all cardinal and intercardinal headings. During the test, readings shall be recorded every 10 minutes. The compass unit error shall not exceed the limit specified in 3.5.4 for this test any time during the test.

4.9.4.1 Tests shall be performed to determine conformance with the normal and rapid settling requirements specified in 3.5.1.1.

4.9.5 Small angle Scorsby test.- The master compass shall be mounted on a Scorsby test stand and shall be settled in accordance with 4.7.4. It shall then be given a 2-hour test on each of two selected readings, with the compass subject to roll, pitch and yaw.

4.9.5.1 The amplitude of roll on each side of the vertical shall be 15 degrees, with a period between 8 and 10 seconds. The amplitude of pitch on each side of the vertical shall be 5 degrees, with a period between 6 and 8 seconds. The amplitude of yaw on each side of the Scorsby test stand reading shall be 5 degrees within a period between 8 and 10 seconds. A reading of the master compass shall be recorded every 30 minutes. To obtain an accurate reading, the motion of the Scorsby test stand shall be stopped, the turntable trammed, and the master compass brought to a vertical position. The master compass error shall not exceed the limit specified in 3.5.4.5 for this test.

4.9.6 Effect of magnetic field test.- The effect of magnetic fields on the compass unit shall be determined by using Helmholtz coils capable of producing north-south, east-west, and vertical fields in turn. The coils shall be capable of producing a magnetic field of five gausses uniform intensity and direction over a cube 3 feet on a side when the permeability of the cube is 1. No adjustment of the field strength shall be made after the insertion of the equipment between the coils. The compass unit shall be mounted midway between the coils with the gyro rotor approximately on the axis of the coils, and shall be settled with no field applied. The field shall be applied in one direction, the compass unit allowed to settle, and the maximum error and new settling position noted. In no case shall the settling with the field applied exceed 4 hours. The direction of the magnetic field shall then be reversed and the procedure repeated. These two tests shall be repeated for each of the two directions of magnetic field perpendicular to the original direction and for a direction of the magnetic field which shall be inclined 45 degrees in the vertical plane parallel to the gyro spin axis. The maximum error shall not exceed the values specified in 3.5.4.5.

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4.9.7 Roll and pitch clearance test.- The mechanical clearances of the compass unit shall be checked under static conditions to determine whether they are in accordance with 3.5.1.5. This shall be effected by settling the compass unit in a level position, tilting it slowly to 45 degrees port roll to check one of the clearances and holding the compass unit in that attitude for 30 minutes, then tilting to 45 degrees starboard roll and holding that attitude for 30 minutes. The compass unit shall not become unsettled dur g this period. The above procedure shall be repeated to check 45 degrees downward and upward pitch.

4.9.8 Large angle Scorsby test.- This test shall be conducted as specified for the small angle Scorsby test, except that the amplitude of roll shall be 40 degrees on each side of the vertical, within a period between 8 and 10 seconds. The amplitude of pitch shall be 10 degrees on ch side of the vertical, with a period between 6 and 8 seconds, and the applied Scorsby yaw shall be omitted. The compass unit error shall not exceed the limit specified in 3.5.4.5.

4.9.9 Shock and vibration.- For the shock and vibration tests, the compass unit shall be settled in accordance with 4.7.5 before the tests. During the vibration test, the compass unit shall be required at all times to not exceed an error of 2 degrees. During the shock test, the compass shall be required at any time not to exceed an error of 5 degrees. After both the shock and vibration tests, the compass unit shall be capable of settling with an error not in excess of the settling point error (1.5 degrees). The compass control unit and power supply unit shall, to meet the shock vibration tests, not suffer a failure which would interfere with proper compass operation during or after the test.

4.9.10 Temperature and humidity test.- For this test the compass shall be settled according to 4.7.4 on one heading. The following procedure shall apply with the compass operating in its normal mode. The compass shall at no time exceed the turntable error of 3.5.4.5 at temperatures for 40° to 110°F. and shall at no time exceed twice the turntable error of 3.5.4.5 for higher temperatures.

Hours	Ambient temperature and relative humidity during test	Angle of tilt
	With the compass at an equilibrium temperature of 40°F., start and settle the compass. This shall require no more than 4 hours. Record the time required.	
0-21	40°F./95 percent relative humidity.	Normal
21-27	Raise the temperature in 6 hours to 149°F., maintain 95 percent relative humidity.	Normal
27-152	Temperature 149°F., relative humidity 95 percent at alternate 1 hour periods, that is, 1 hour at 95 percent relative humidity and the next hour with relative humidity uncontrolled. Measure air temperature external and internal to the enclosure and check parts for hot spots, at the end of 125 hours.	Normal
152-156	Lower the temperature and humidity to room conditions, nominally 75°F. and 50 to 60 percent relative humidity.	Normal

Throughout the above test compass error shall be recorded either continuously or at regular 15-minute intervals. Hot spot temperatures shall not exceed the maximum dictated by affected insulation as specified in MIL-I-983 and shall be performed according to MIL-I-983. The maximum internal temperature rise shall be recorded and shall not exceed 90°F. from the ambient. Equipment installation and the air change rate capability of the test chamber shall be in accordance with MIL-I-983.

4.9.10.1 Extended storage.- For this test the procedure shall be in accordance with MIL-I-983. The equipment after stabilizing at the room temperature shall be capable of settling with an error no greater than the settling point error of 3.5.4.5.

4.9.10.2 Extreme temperature for exposed location.- The compass shall be settled in accordance with 4.7.4 and shall be operated on one heading with readings taken as specified for the turntable test (see 4.9.4). The compass unit error shall not exceed twice the turntable error at any time during the test.

4.9.10.3 Power failure.- A test shall be performed to determine conformance with 3.5.1.14.

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4.9.10.4 The compass unit shall be settled. The mounting table shall be turned in azimith smoothly from rest to a velocity of 100 degrees per second for approximately one revolution and then smoothly decelerated to rest. The overall lag of the compass during the constant velocity part of the motion shall not exceed 15 minutes of arc. The compass unit error measured at 5 minute intervals for 30 minutes after the rotation shall not exceed 3 degrees.

4.9.10.5 In addition to the specified environmental tests, the contractor should demonstrate the equipment reliability by a "burn in" test applied for minimum of 1000 hours on 3 seperate preproduction models of the compass. All production models of the equipment shall require a minimum of 100 hours of "burn in" time in addition to the quality conformance inspection. During "burn in" the equipment shall be operated with an "on" (settling) cycle of approximately 20 hours duration and an "off" cycle of approximately 2 hours. The recorded data shall consist of a log of failures and component replacement. The data shall also indicate the time required for settling as well as the compass performance accuracy during its "on" cycle.

4.10 Failures.- The occurence of any of the following conditions shall require the subsequent running of the turntable test, small angle Scorsby test, and such other parts of the quality conformance inspection as the laboratory may consider to be affected by the failure or the subsequent adjustment:

(a) Compass unit error in excess of the allowable limits during the turntable test or small angle Scorsby test.
(b) Failure to settle in accordance with 4.7.4 at any time during the quality conformance inspection.
(c) Failure, defective operation, breakdown on the insulation test, or need for adjustment of any part of the sensitive element or follow-up system, including amplifier, at any time during the quality conformance inspection.

4.10.1 A failure of any auxiliary unit of the equipment (except parts of the sensitive element and follow-up system) which affects the operation of the master compass, while being tested as specified in 4.5 shall require the rerunning of the part of the quality conformance inspection on which the failure occurred.

4.11 Trial rerun.- In case the master compass error exceeds the allowed limit during the turntable test or small angle or large angle Scorsby test, and there is no apparent cause for this condition, contractor may request a rerun of the part of the test on which the excessive error was observed for the purpose of determining the cause of the error. If the reasons for such a run are satisfactory to the laboratory, the rerun shall be made under carefully observed conditions. If the results of the rerun are satisfactory, the part of the quality conformance inspection in question shall be considered complete, and the excessive error during the original run shall not be considered a failure.

4.12 Material examination after test.- The gyro compass equipment, after completing successfully the entire quality conformance inspection, shall be subjected to material inspection. Any parts found defective or showing undue wear shall be replaced by the contractor at his expense, and so much of gyro compass equipment as the Government inspector may deem to be affected by replacement shall again be subjected to all applicable parts of the quality conformance inspection under the same conditions and stipulations as newly submitted equipment.

4.13 Inspection of preparation for delivery.- The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

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5.1.1 Gyro compass, vehicle.-

5.1.1.1 Preservation and packaging.- Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged by level A; packed by level C and marked by levels A and C, respectively, in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing and marking.- The equipment and accessories, repair parts, and technical publications shall be preserved and packaged by level A or C, packed by level A or B as specified, and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Whether preproduction sample is required (see 3.1 and 4.3).
(d) Whether power converter unit is required (see 3.2).
(e) Inventory control point for repair parts.
(f) Quantity of maintenance parts.
(g) Quantity of manuals.
(h) Quantity of compass units to be supplied by the manufacturer complete with synchro transmitter.
(i) Frequency (60 or 400 cycles) of synchro transmitter.
(j) Preservation and packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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